

Lecture 4 - Petrinets

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- Petrinets are used to represent concurrent systems.
 - Petrinets are bipartite graphs.
 - An alternative representation to LTSs.
- There exist different kinds of Petrinets.
- An *elementary* Petrinet is composed of the following elements:
 1. *Places*.
 - Places represent a resource (essentially a state) that the system can be in.
 - Depicted by a circle.
 2. *Transitions*.
 - Transitions represent an action that a system performs.
 - Depicted by a rectangle.
 3. *Arcs*.
 - Arcs represent directed connections between places and transitions.
 - * Since Petrinets are bipartite, arcs cannot connect places to places or transitions to transitions.
 - Depicted by an arrow
 4. *Tokens*.
 - Tokens represent the current state of the system.
 - Tokens exist within places and move between places according to the arcs connecting places to transitions.
 - A petrinet can have multiple tokens simultaneously.
 - The current position of tokens in the system is depicted by a dot in the place the token resides in.
- Relative to a given transition:
 - An *input* place is a place which connects to that transition.
 - An *output* place is a place which that transition connects to.
- For a transition to fire, all input places to that transition must have a token inside it.
- Multiple transitions satisfying this condition can fire at this same time.
 - This is how Petrinets model the simultaneous execution of an action.
- When a transition fires, all output places are populated with a token and all input places are depopulated of their tokens.
- A given place can be both an input and output place of a transition.

- A *deadlock* describes a situation where tokens are arranged in such a way that there exists no valid transition for any tokens to take.
- The *firing sequence* of an elementary Petrinet are sequences of valid transitions that can be taken based on the initial location of tokens.
- The Petrinet representation of an LTS representing a concurrent system can be attained by representing the processes in parallel composition as Petrinets and “glueing” together shared transitions.
- *Labelled elementary Petrinets* are essentially the same as elementary Petrinets, with the distinction being that if two transitions can be fired simultaneously, they must have distinct labels.
- *Reachability graphs* are finite state machines used to represent the behaviour of Petrinets in LTS form.
 - For a given FSP, the reachability graph of the FSP’s Petrinet representation is equivalent to the FSP’s LTS representation.
 - Reachability graphs may allow for actions to be executed simultaneously.

Ex. Represent the following FSPs:

$\text{MAKER} = \text{make} \rightarrow \text{ready} \rightarrow \text{MAKER}$
 $\text{USER} = \text{ready} \rightarrow \text{user} \rightarrow \text{USER}$
 $||\text{MAKER_USER} = \text{MAKER} || \text{USER}$

(a) As LTSs.

Solution:

The LTSs for these FSPs are given by:

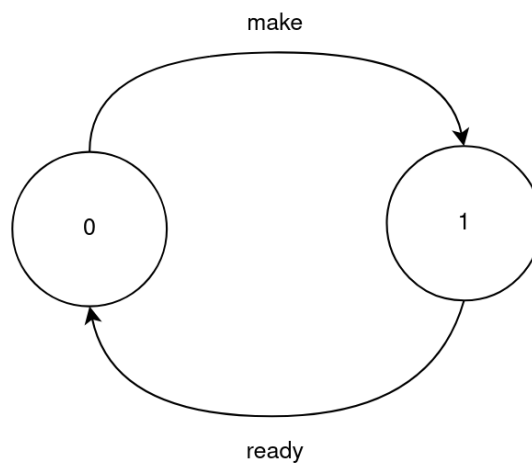


Figure 1: LTS representation of **MAKER**.

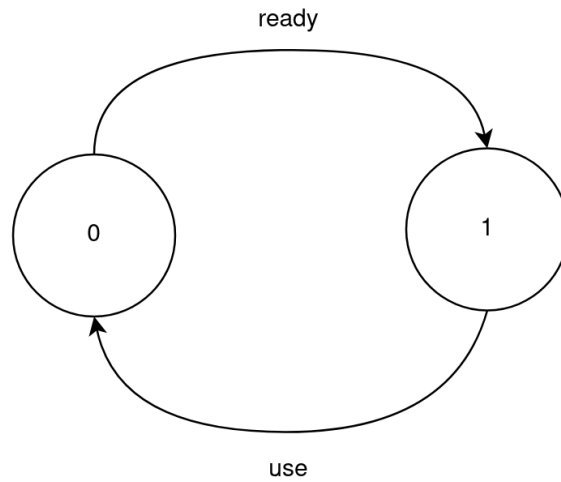


Figure 2: LTS representation of USER.

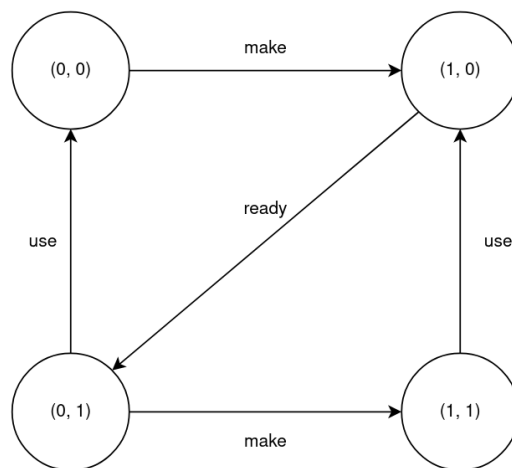


Figure 3: LTS representation of MAKER.USER.

(b) As Petrinets.

Solution:

The Petrinets for these FSPs are given by:

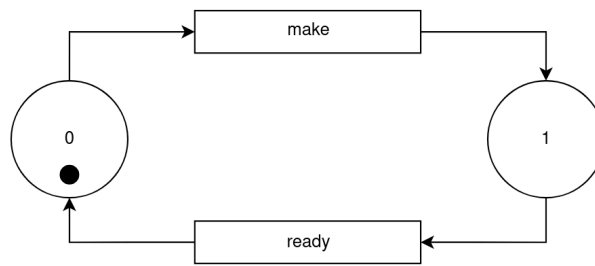


Figure 4: Petrinet representation of MAKER.

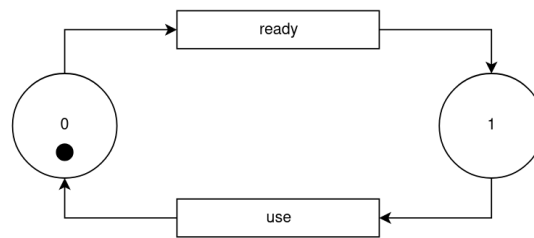


Figure 5: Petrinet representation of USER.

By glueing these two Petrinets by the common transition “**ready**”, we can easily represent the Petrinet for MAKER_USER as:

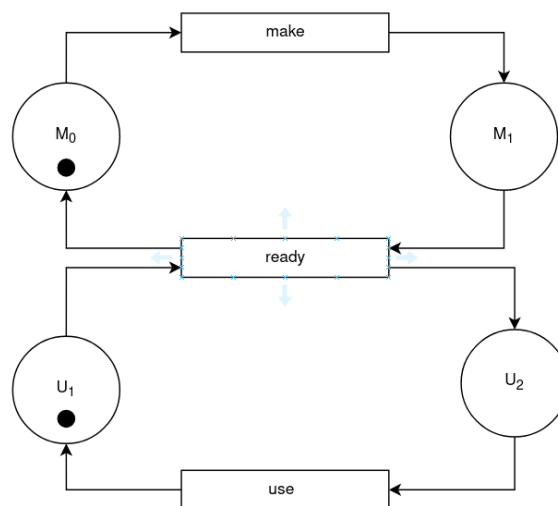


Figure 6: Petrinet representation of MAKER_USER.

Ex. Construct the reachability graph for the Petrinet representation of the MAKER_USER.

FSP.

Solution:

The reachability graph for the Petrinet representation of `MAKER_USER` is given by:

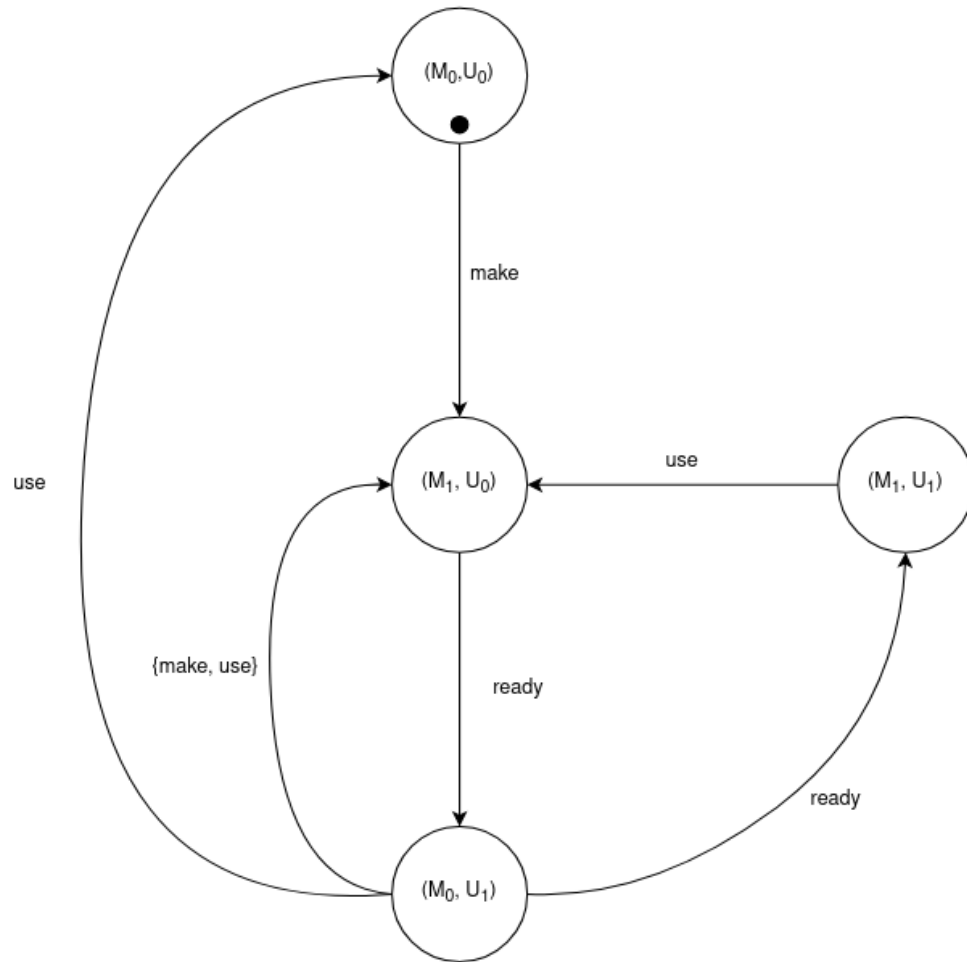


Figure 7: Reachability graph of `MAKER_USER` Petrinet.